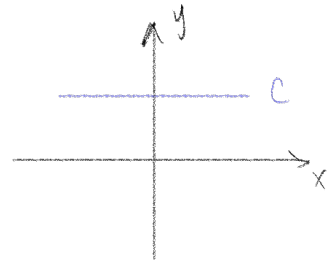


Simple Graphs

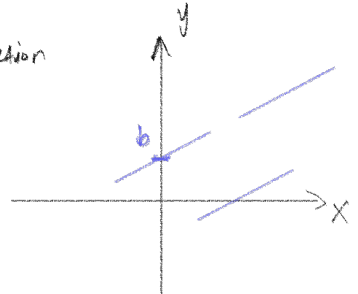
① $f(x) = c \iff f := \{(a, b) : f(a) = b\}$
 $= \{(x, c) : x \in \mathbb{R}\}$



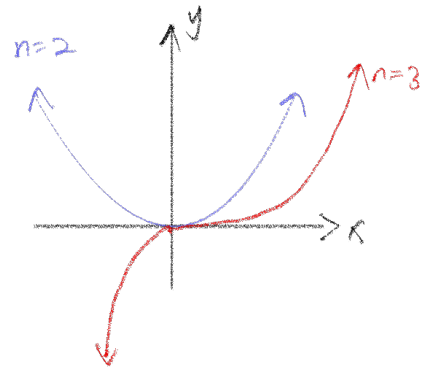
② $f(x) = mx + b$ linear function

The reason is not the correct graph is b/c. the function is continuous.

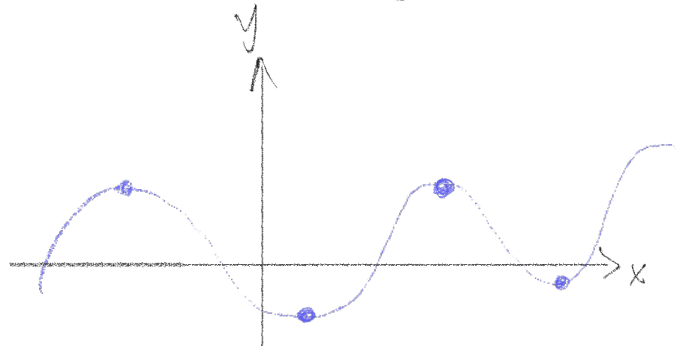
Continuous at $x = a \rightarrow \lim_{x \rightarrow a} f(x) = f(a).$



③ $f(x) = x^n$ Power function



④ Polynomial: $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$



Pathological examples

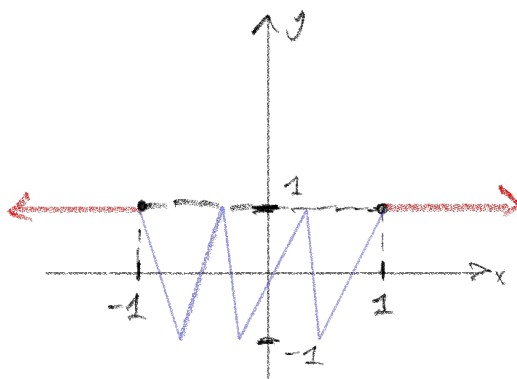
Def. A piecewise function is defined on the interval of $\mathbb{R}$.

($\amalg$: disjoint union) $A \amalg B$ assumes that $A \cap B = \emptyset$

if $\mathbb{R} = I_1 \amalg I_2 \amalg \dots \amalg I_n$

Let f be defined as

$$f(x) = \begin{cases} 1 & \text{if } |x| \geq 1 \\ f(\frac{1}{n}) = (-1)^{n+1} & (n \in \mathbb{Z}_{>0}) \\ f(\frac{1}{n}) = (-1)^{n+1} & (n \in \mathbb{Z}_{<0}) \end{cases}$$

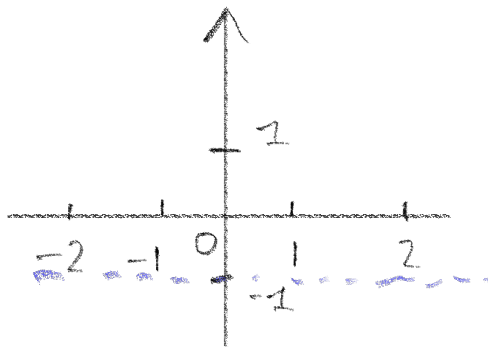


Variant: $f(x) = \sin(\frac{1}{x})$

Q: Is f continuous at $x=0$?

② Let $g(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ -1 & \text{if } x \notin \mathbb{Q} \end{cases}$

Not continuous anywhere



$$(3) \quad h(x) = \begin{cases} 1/q & \text{if } x = p/q, x \in \mathbb{Q}, \text{ in lowest terms} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases}$$

Continuous at irrational # ($x \notin \mathbb{Q}$)

Not continuous at rational # ($x \in \mathbb{Q}$)

$$(4) \quad C: [0, 1] \rightarrow [0, 1]$$

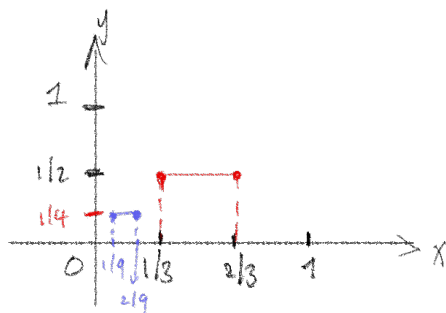
Process: Trisect $[0, 1]$

Assign $1/2$ to $[1/3, 2/3]$

Repeat Trisect $[0, 1/3]$, assign $1/4$ to $[1/9, 2/9]$.

Trisect $[2/3, 1]$, assign $3/4$ to $[7/9, 8/9]$ etc.

Cantor Function



Properties

- Continuous on all of $[0, 1]$.
- non-decreasing (every value b/w. 0 and 1 is achieved somewhere)
- Derivative is 0 almost everywhere.
- Arc Length of this graph is 2.

Implicit Function

- $f := \{(x, y) \in \mathbb{R}^2 : (x-a)^2 + (y-b)^2 = r^2\}$ is an example of implicit function. ^(y ≥ b)

Set of (x, y) that are distance r away from (a, b) .

- WLOG

let $P = (c, 0)$, $Q = (-c, 0)$.

What is $\{R \in \mathbb{R}^2 : |PR| + |RQ| = 2a\}$?

